Agile Software Development and DevOps Lab

Experiment 1

Aim: To perform version control operations using Git.

LO mapped: LO 2

LO: Use Git effectively for version control, source code management, and collaborative team development.

Theory:

Version Control

Version control is a system that records changes made to files over time, allowing users to track modifications, compare different versions, restore previous versions, and collaborate efficiently. Git is a distributed version control system that maintains the complete history of a project, making it easy to trace when changes were made, who made them, and why.

Commands:

git status
Purpose: Displays the current state of the working directory, including modified, staged, and untracked files before they are committed.

git add <file_name>
Purpose: Adds changes from the working directory to the staging area, preparing them to be included in the next version (commit).

git commit -m "Commit message"
Purpose: Records the staged changes as a new version in the repository along with a descriptive commit message.

git log
Purpose: Displays the complete history of commits, allowing users to trace the evolution of the project and identify when specific changes were made.

git show <commit_id>
Purpose: Displays detailed information about a specific commit, including the author, date, commit message, and the changes introduced in that version.

git diff
Purpose: Compares differences between file versions, staged changes, or commits, helping users analyze modifications before or after committing them.

git blame <file_name>
Purpose: Shows who last modified each line of a file and the commit in which the change was made, making it useful for tracing the history of individual lines of code.

git tag <tag_name>
Purpose: Marks a specific commit with a meaningful label, such as a software release or milestone, making important versions easy to identify.

Authorization

Authorization is the process of granting or restricting access to a Git repository. It allows repository owners to control who can view, clone, push, or manage the project. Authorization is typically managed on remote Git hosting platforms such as GitHub or GitLab by adding or removing collaborators and assigning appropriate permissions.

Commands:

git clone <repository_url>
Purpose: Allows an authorized user to create a local copy of a repository to which they have access.

git pull origin <branch_name>
Purpose: Retrieves the latest changes from a remote repository. This command works only if the user has permission to access the repository.

git push origin <branch_name>
Purpose: Uploads local commits to the remote repository. The operation is successful only if the user has write access.

git remote add origin <repository_url>
Purpose: Connects a local repository to a remote repository that the user is authorized to access.

Granting and Revoking Access

Git itself does not provide commands to grant or revoke access to other users. These permissions are managed through the repository hosting service.

- Grant Access: Add a collaborator or team member through the repository settings (e.g., GitHub's *Manage Access* feature).
- Revoke Access: Remove the collaborator or change their permission level through the repository settings.

Rollback

Rollback is the process of undoing changes or returning a project to a previous stable state. Git provides several commands to discard local changes, undo commits, or restore earlier versions while maintaining project history.

Commands

git restore <file_name>
Purpose: Restores a modified file to its last committed version, discarding local changes.

git reset <commit_id>
Purpose: Moves the current branch to a previous commit and can also unstage changes.

git reset --hard <commit_id>
Purpose: Removes all commits and uncommitted changes made after the specified commit.

git revert <commit_id>
Purpose: Creates a new commit that reverses the changes introduced by a previous commit without deleting project history.

Revoking

Revoking in Git refers to removing or modifying access or connections to remote repositories. It is commonly used when a remote repository is no longer required or when its configuration needs to be changed.

Commands

git remote remove origin
Purpose: Removes the connection between the local repository and the specified remote repository.

git remote rename origin upstream
Purpose: Renames an existing remote repository to a new name.

git remote -v
Purpose: Displays all configured remote repositories and their associated URLs.

Output:

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git log
commit 223ed75375cd572ff26690dc0f9c0f479832840e (HEAD -> master)
Author: Soham Ajwani <sohamajwani03@gmail.com>
Date:   Tue Jul 14 21:08:22 2026 +0530

    initial version with addition fn

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git diff
diff --git a/calculator.py b/calculator.py
index e69de29..6117f70 100644
--- a/calculator.py
+++ b/calculator.py
@@ -0,0 +1,9 @@
+def add(a, b):
+    return a + b
+
+def subtract(a, b):
+    return a - b
+
+
+print("Addition:", add(10, 5))
+print("Subtraction:", subtract(10, 5))
\ No newline at end of file
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git push origin master
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 549 bytes | 549.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/sohamajwani/ASD_T1_04.git
 * [new branch]      master -> master
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git show bb2b7721e30a239bc9248bfaed56abf40b5a001f
commit bb2b7721e30a239bc9248bfaed56abf40b5a001f (HEAD -> master, origin/master)
Author: Soham Ajwani <sohamajwani03@gmail.com>
Date:   Tue Jul 14 21:10:45 2026 +0530

    added subtraction function

diff --git a/calculator.py b/calculator.py
index e69de29..6117f70 100644
--- a/calculator.py
+++ b/calculator.py
@@ -0,0 +1,9 @@
+def add(a, b):
+    return a + b
+
+def subtract(a, b):
+    return a - b
+
+
+print("Addition:", add(10, 5))
+print("Subtraction:", subtract(10, 5))
\ No newline at end of file
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git push --set-upstream origin master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 322 bytes | 322.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/sohamajwani/ASD_T1_04.git
   bb2b772..e65143d  master -> master
branch 'master' set up to track 'origin/master'.
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git blame calculator.py
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  1) def add(a, b):
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  2)     return a + b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  3)
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  4) def subtract(a, b):
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  5)     return a - b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  6)
e65143d5 (Soham Ajwani 2026-07-14 21:15:07 +0530  7) def multiply(a, b):
e65143d5 (Soham Ajwani 2026-07-14 21:15:07 +0530  8)     return a * b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  9)
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530 10) print("Addition:", add(10, 5))
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530 11) print("Subtraction:", subtract(10, 5))

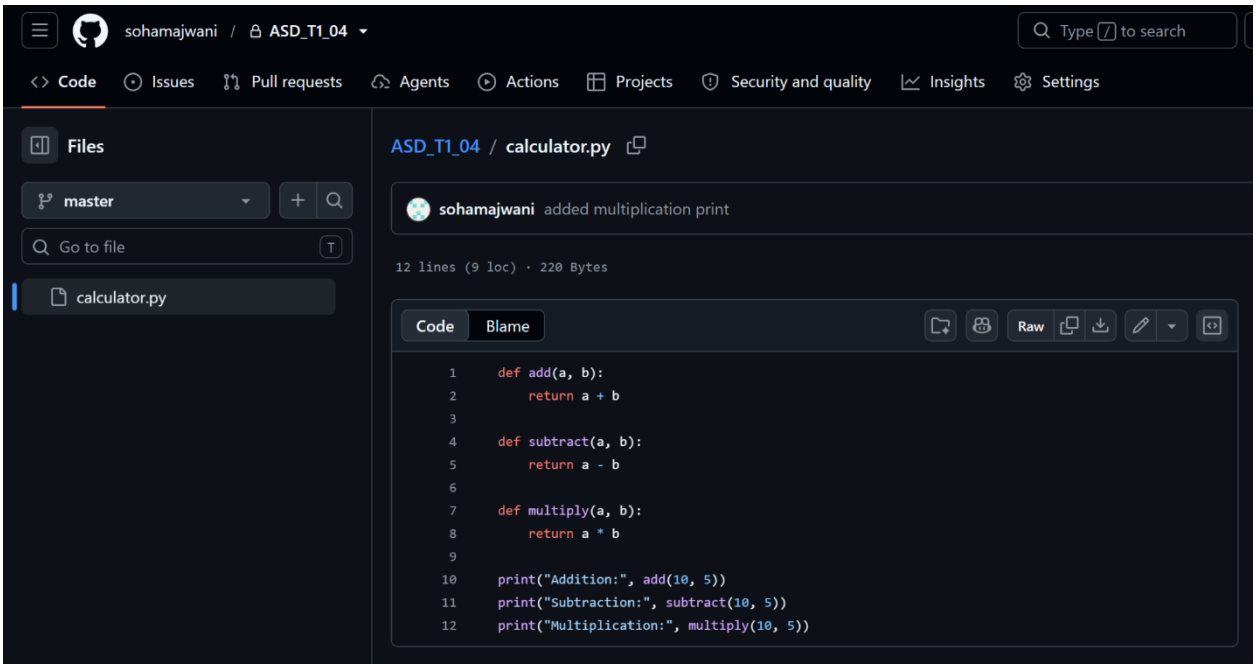
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git tag v1.0

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git tag
v1.0
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git blame calculator.py
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  1) def add(a, b):
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  2)     return a + b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  3)
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  4) def subtract(a, b):
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  5)     return a - b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  6)
e65143d5 (Soham Ajwani 2026-07-14 21:15:07 +0530  7) def multiply(a, b):
e65143d5 (Soham Ajwani 2026-07-14 21:15:07 +0530  8)     return a * b
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530  9)
bb2b7721 (Soham Ajwani 2026-07-14 21:10:45 +0530 10) print("Addition:", add(10, 5))
55b4fb6f (Soham Ajwani 2026-07-14 21:20:00 +0530 11) print("Subtraction:", subtract(10, 5))
55b4fb6f (Soham Ajwani 2026-07-14 21:20:00 +0530 12) print("Multiplication:", multiply(10, 5))

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git tag v2.0

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git tag
v1.0
v2.0
```



Authorization

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15 (master)
$ git clone https://github.com/SaniyaBijlani/ASD_T1_15.git
Cloning into 'ASD_T1_15'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 9 (delta 1), reused 9 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
Resolving deltas: 100% (1/1), done.
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git add calculator.py

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git commit -m "added division function"
[master b19d489] added division function
 1 file changed, 7 insertions(+), 1 deletion(-)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git diff

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 378 bytes | 378.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/SaniyaBijlani/ASD_T1_15.git
   f0687a1..b19d489  master -> master
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 312 bytes | 34.00 KiB/s, done.
From https://github.com/SaniyaBijlani/ASD_T1_15
 * branch            master     -> FETCH_HEAD
   b19d489..eaa3e70  master     -> origin/master
Updating b19d489..eaa3e70
Fast-forward
 calculator.py | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 312 bytes | 34.00 KiB/s, done.
From https://github.com/SaniyaBijlani/ASD_T1_15
 * branch            master     -> FETCH_HEAD
   b19d489..eaa3e70  master     -> origin/master
Updating b19d489..eaa3e70
Fast-forward
 calculator.py | 3 ++-
 1 file changed, 2 insertions(+), 1 deletion(-)
```

```
Saniya@LAPTOP-70TMSUFS MINGW64 ~/OneDrive/Desktop/college_work/ASD/ASD_T1_15 (master)
$ git pull origin master
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (1/1), done.
remote: Total 3 (delta 1), reused 3 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 358 bytes | 21.00 KiB/s, done.
From https://github.com/SaniyaBijlani/ASD_T1_15
 * branch            master     -> FETCH_HEAD
   f0687a1..b19d489  master     -> origin/master
Updating f0687a1..b19d489
Fast-forward
 calculator.py | 8 +++++++-
 1 file changed, 7 insertions(+), 1 deletion(-)
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git add calculator.py

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git commit -m "last print statement"
[master f97163c] last print statement
 1 file changed, 2 insertions(+), 1 deletion(-)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_15/ASD_T1_15 (master)
$ git push
remote: Permission to SaniyaBijlani/ASD_T1_15.git denied to sohamajwani.
fatal: unable to access 'https://github.com/SaniyaBijlani/ASD_T1_15.git/': The requested URL returne
d error: 403
```

Rollback

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git branch backup-before-rollback

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git log --oneline
55b4fb6 (HEAD -> master, tag: v2.0, origin/master, backup-before-rollback) added multiplication print
e65143d (tag: v1.0) added multiplication fn
bb2b772 added subtraction function
223ed75 initial version with addition fn

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git status
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   calculator.py

no changes added to commit (use "git add" and/or "git commit -a")

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git restore calculator.py
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git add calculator.py

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git commit -m "added division function"
On branch master
Your branch is up to date with 'origin/master'.

nothing to commit, working tree clean

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git log --oneline
55b4fb6 (HEAD -> master, tag: v2.0, origin/master, backup-before-rollback) added multiplication print
e65143d (tag: v1.0) added multiplication fn
bb2b772 added subtraction function
223ed75 initial version with addition fn

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git reset HEAD~1
Unstaged changes after reset:
M       calculator.py
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git status
On branch master
Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   calculator.py

no changes added to commit (use "git add" and/or "git commit -a")

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ ^C

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git add calculator.py

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git commit -m "added division again"
[master f1c2a37] added division again
 1 file changed, 2 insertions(+), 1 deletion(-)
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git reset --hard HEAD~1
HEAD is now at e65143d added multiplication fn

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git log --oneline
e65143d (HEAD -> master, tag: v1.0) added multiplication fn
bb2b772 added subtraction function
223ed75 initial version with addition fn

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ ^C

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git add calculator.py

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git commit -m "Added division"
On branch master
Your branch is behind 'origin/master' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
```

```
[master 0265ff2] Revert "added division function"
 1 file changed, 1 insertion(+), 6 deletions(-)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git log --oneline
0265ff2 (HEAD -> master) Revert "added division function"
1c6aba8 added division function
e65143d (tag: v1.0) added multiplication fn
bb2b772 added subtraction function
223ed75 initial version with addition fn
```

Revoking

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v
origin  https://github.com/sohamajwani/ASD_T1_04.git (fetch)
origin  https://github.com/sohamajwani/ASD_T1_04.git (push)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v
origin  https://github.com/sohamajwani/ASD_T1_04.git (fetch)
origin  https://github.com/sohamajwani/ASD_T1_04.git (push)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote rename origin upstream

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v
upstream        https://github.com/sohamajwani/ASD_T1_04.git (fetch)
upstream        https://github.com/sohamajwani/ASD_T1_04.git (push)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote rename upstream origin

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v
origin  https://github.com/sohamajwani/ASD_T1_04.git (fetch)
origin  https://github.com/sohamajwani/ASD_T1_04.git (push)

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
```

```
soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote remove origin



soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote add origin https://github.com/sohamajwani/ASD_T1_04.git

soham@DESKTOP-6BV2HJL MINGW64 ~/Documents/ASD_T1_04 (master)
$ git remote -v
origin  https://github.com/sohamajwani/ASD_T1_04.git (fetch)
origin  https://github.com/sohamajwani/ASD_T1_04.git (push)
```

Conclusion: Through this experiment, version control operations using Git were successfully performed, demonstrating its effectiveness in tracking file modifications and tracing project history. Additionally, key operations for remote authorization, safety rollbacks, and managing remote repository connections were executed. This satisfies the learning objective of utilizing Git for source code management and collaborative team development.

LO2 mapped successfully.